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Lab 3. Creating and Managing Schema Objects

1. Create the following tables:

--DEPT table:

```
CREATE TABLE DEPT(
   DEPT_NO NUMBER(2) Primary key,
   DEPT_NAME VARCHAR2(20) unique,
   HOD VARCHAR2(20)
);
```

--EMPLOYEE table:

```
CREATE TABLE EMPLOYEE (
   EMP_NO NUMBER(5) PRIMARY KEY,
   EMP_NAME VARCHAR2(20),
   SAL NUMBER(6) CHECK (SAL >= 1000),
   DEPT_NO NUMBER(2),
   DJ DATE,
   DEG VARCHAR2(20),
        CONSTRAINT FK_DEPT_NO FOREIGN KEY (DEPT_NO) REFERENCES
DEPT(DEPT_NO)
);
```

--LEAVE table:

```
CREATE TABLE LEAVE (
   LEAVE_TYPE CHAR(1) PRIMARY KEY,
   LEAVE_NAME VARCHAR2(20),
   NO_LEAVES NUMBER(2) CHECK (NO_LEAVES <= 20)
);
```

--EMP_LEAVE table:

```
CREATE TABLE EMP_LEAVE (
   EMP_NO NUMBER(5),
   LEAVE_TYPE CHAR(1) NOT NULL,
   START_DATE DATE,
   END_DATE DATE ,
   PRIMARY KEY (EMP_NO, START_DATE),
   CONSTRAINT CHECK_END_DATE CHECK (END_DATE >= START_DATE),
         CONSTRAINT FK_EMP_NO FOREIGN KEY (EMP_NO) REFERENCES
EMPLOYEE(EMP_NO),
      CONSTRAINT FK_LEAVE_TYPE FOREIGN KEY (LEAVE_TYPE) REFERENCES
LEAVE(LEAVE_TYPE)
);
```

2. Add data to the above tables:

--DEPT table:

```
INSERT INTO DEPT (dept_no, dept_name, hod) VALUES (1, 'MAINFRAME', 'GEORGE');
```

INSERT INTO DEPT (DEPT_NO, DEPT_NAME, HOD) VALUES (2, 'CLIENT/SERVER', 'BILL');
INSERT INTO DEPT (DEPT_NO, DEPT_NAME, HOD) VALUES (3, 'SYSTEMS', 'GARRY');
INSERT INTO DEPT (DEPT_NO, DEPT_NAME, HOD) VALUES (4, 'INTERNET', 'PAUL');
INSERT INTO DEPT (DEPT_NO, DEPT_NAME, HOD) VALUES (5, 'ACCOUNTS', 'ANDY');

--EMPLOYEE table:
INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (101, 'GEORGE', 12000, 1, TO_DATE('12-JUL-2001', 'DD-MON-YYYY'), 'PM');

INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (102, 'BILL', 12000, 2, TO_DATE('14-JUL-2001', 'DD-MON-YYYY'), 'PM');

INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (103, 'GARRY', 15000, 3, TO_DATE('01-JUL-2001', 'DD-MON-YYYY'), 'PM');

INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (104, 'PAUL', 11000, 4, TO_DATE('02-JUL-2001', 'DD-MON-YYYY'), 'PL');

INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (105, 'ANDY', 7000, 5, TO_DATE('25-JUN-2001', 'DD-MON-YYYY'), 'AM');

INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (106, 'KEATS', 10000, 1, TO_DATE('17-JUL-2001', 'DD-MON-YYYY'), 'SA');

INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (107, 'JOEL', 8000, 2, TO_DATE('15-JUL-2001', 'DD-MON-YYYY'), 'SP');

INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (108, 'ROBERTS', 7500, 2, TO_DATE('15-JUL-2001', 'DD-MON-YYYY'), 'PRO');

INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (109, 'HERBERT', 8000, 4, TO_DATE('22-JUL-2001', 'DD-MON-YYYY'), 'SA');

INSERT INTO EMPLOYEE (EMP_NO, EMP_NAME, SAL, DEPT_NO, DJ, DEG)
VALUES (110, 'MICHEAL', 6000, 4, TO_DATE('15-JUL-2001', 'DD-MON-YYYY'), 'PRO');

--LEAVE table:
INSERT INTO LEAVE (LEAVE_TYPE, LEAVE_NAME, NO_LEAVES)
VALUES ('C', 'CASUAL', 15);

INSERT INTO LEAVE (LEAVE_TYPE, LEAVE_NAME, NO_LEAVES)
VALUES ('E', 'EARNING', 5);

INSERT INTO LEAVE (LEAVE_TYPE, LEAVE_NAME, NO_LEAVES)
VALUES ('O', 'OVERTIME', 5);

INSERT INTO LEAVE (LEAVE_TYPE, LEAVE_NAME, NO_LEAVES)

```sql
VALUES ('S', 'SICK', 15);
```

--EMP_LEAVES table:

```sql
INSERT INTO EMP_LEAVE (EMP_NO, LEAVE_TYPE, START_DATE, END_DATE)
VALUES (102, 'S', TO_DATE('23-JUL-2001', 'DD-MON-YYYY'), TO_DATE('25-JUL-2001',
'DD-MON-YYYY'));

INSERT INTO EMP_LEAVE (EMP_NO, LEAVE_TYPE, START_DATE, END_DATE)
VALUES (104, 'C', TO_DATE('24-JUL-2001', 'DD-MON-YYYY'), TO_DATE('25-JUL-2001',
'DD-MON-YYYY'));

INSERT INTO EMP_LEAVE (EMP_NO, LEAVE_TYPE, START_DATE, END_DATE)
VALUES (104, 'S', TO_DATE('28-JUL-2001', 'DD-MON-YYYY'), TO_DATE('29-JUL-2001',
'DD-MON-YYYY'));

INSERT INTO EMP_LEAVE (EMP_NO, LEAVE_TYPE, START_DATE, END_DATE)
VALUES (101, 'C', TO_DATE('27-JUL-2001', 'DD-MON-YYYY'), TO_DATE('28-JUL-2001',
'DD-MON-YYYY'));

INSERT INTO EMP_LEAVE (EMP_NO, LEAVE_TYPE, START_DATE, END_DATE)
VALUES (106, 'O', TO_DATE('28-JUL-2001', 'DD-MON-YYYY'), TO_DATE('29-JUL-2001',
'DD-MON-YYYY'));

INSERT INTO EMP_LEAVE (EMP_NO, LEAVE_TYPE, START_DATE, END_DATE)
VALUES (109, 'C', TO_DATE('01-AUG-2001', 'DD-MON-YYYY'), TO_DATE('02-AUG-2001',
'DD-MON-YYYY'));

INSERT INTO EMP_LEAVE (EMP_NO, LEAVE_TYPE, START_DATE, END_DATE)
VALUES (103, 'C', TO_DATE('02-AUG-2001', 'DD-MON-YYYY'), TO_DATE('05-AUG-2001',
'DD-MON-YYYY'));

INSERT INTO EMP_LEAVE (EMP_NO, LEAVE_TYPE, START_DATE, END_DATE)
VALUES (105, 'S', TO_DATE('17-AUG-2001', 'DD-MON-YYYY'), NULL);

INSERT INTO EMP_LEAVE (EMP_NO, LEAVE_TYPE, START_DATE, END_DATE)
VALUES (108, 'S', TO_DATE('23-AUG-2001', 'DD-MON-YYYY'), NULL);
```

3. Write queries related to Employees Management Application:

a. Display employees who have joined in the last 15 days.

--A

```sql
SELECT * FROM employee
WHERE (TO_DATE('28-JUL-2001', 'DD-MON-YYYY') - DJ) <= 15;
```

b. Display the details of employees who have taken more than 10 sick leave days or more than 15 leave days.

--B

```sql
SELECT e.emp_no, e.emp_name
FROM employee e
```

```
JOIN emp_leave el ON el.EMP_NO = e.emp_no
JOIN leave l ON el.leave_type = l.leave_type
WHERE l.leave_name = 'SICK'
GROUP BY e.emp_no, e.emp_name
HAVING SUM(el.END_DATE - el.START_DATE + 1) > 10

UNION

SELECT e.emp_no, e.emp_name
FROM employee e
JOIN emp_leave el ON el.EMP_NO = e.emp_no
JOIN leave l ON el.leave_type = l.leave_type
GROUP BY e.emp_no, e.emp_name
HAVING SUM(el.END_DATE - el.START_DATE + 1) > 15;
```

c. Change the deptno of employee 104 to deptno of the 'Internet' dept.

--C

```
UPDATE EMPLOYEE
SET dept_no = (SELECT DEPT_NO FROM dept WHERE DEPT_NAME = 'INTERNET')
WHERE emp_no = 104;
```

d. Delete details of leaves taken by employees whose empno is the highest empno.

--D

```
DELETE FROM emp_leave
WHERE emp_no = (SELECT MAX(EMP_NO) FROM employee);
```

4. Write an SQL statement to create a view named vw_ManyLeavesEmp that displays the empno of the employee who has taken more than 2 leaves in the current month.

```
CREATE OR REPLACE VIEW vw_ManyLeavesEmp AS
SELECT EMP_NO
FROM EMP_LEAVE
WHERE EXTRACT(MONTH FROM START_DATE)= EXTRACT(MONTH FROM SYSDATE)
  AND EXTRACT(YEAR FROM START_DATE) = EXTRACT(YEAR FROM SYSDATE)
GROUP BY EMP_NO
HAVING COUNT(*) > 2;
```

5. Write an SQL statement to create a view named vw_LeavesPriodMonth that displays details of leaves where the leave started in the previous month and is not yet completed.

```
CREATE VIEW vw_LeavesPeriodMonth AS
SELECT *
FROM EMP_LEAVE
WHERE EXTRACT(MONTH FROM START_DATE)=
                    (EXTRACT(MONTH FROM SYSDATE) -1)
  AND EXTRACT(YEAR FROM START_DATE) = EXTRACT(YEAR FROM SYSDATE)
```

```
AND (END_DATE IS NULL);
```